FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	534275
Award Program	Developing Massively Parallel Sequence for Agriculture Surveillance
Project Title	
Grant Start Date	07/01/2017
Grant End Date	06/30/2021
Total Grant Budget	$339,969.00
Total FFAR Budget	$169,960.00
Matching Funder(s)	
Final Report Date	04/13/2023
Project Director/Principal Investigator Information	
Full Name	Jonas King
Email	Not sure where to contact him now
Phone	
Grantee Organization Information	
Organization Name	Mississippi State University
Address	32 Creelman St.
City, State Zip	Mississippi State, MS
Tax ID	640410581

FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

Authorized Signing Official (ASO) Information	
Name	
ASO Title	
Email	
Phone	

General Information

1.1. Please list the geographic location(s) – city, state, congressional district - wherethe work was conducted. If the work was conducted outside of the US, please list the city and country.

Mississippi State, MS MS-003

1.2. How many new jobs were created because of this grant funding?

Three Graduate Students were supported in full or in part by this funding, and three undergraduate student worker were supported by this funding

1.3. How many jobs were maintained because of this grant funding?

One Graduate student was able to graduate while being supported by this funding.

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profitstatus since you were awarded the grant? If yes, please explain.

No Change

1.5. Has your organization undergone a recent name change? If so, please providethe new name of your organization.

No Change

2. Scientific Report

2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

The number of traditional systematists is declining, the scale of global commerce has concurrently exploded, and introductions of exotic plant pathogens and insect pests have reached unprecedented levels. These factors point to the pressing need for the development of a reliable toolkit for molecular surveillance of introduced pests and pathogens. Targeted methods such as PCR and metabarcoding require prior knowledge or expectations of the target pathogen. Nontargeted metagenomic analyses via massively parallel sequencing (MPS, also known as next-generation sequencing) offers a means of pathogen detection that works independently of any prior knowledge about a pathogen and can indeed be used for the discovery and characterization of novel pathogens. Several hurdles, most related to costs and computational requirements, remain to be surmounted before metagenomics becomes a viable means of pathogen and pest surveillance. The ability to employ these methods in the field will also not become a reality until several additional hurdles, related to portability and computational bottlenecks, are overcome. Here, we use an unbiased MPS approach to identify known and novel pests and pathogens of agricultural and societal importance in one assay. We demonstrate the capabilities of MPS for use in agriculture as a surveillance tool for potentially harmful insects and pathogens. Routine use of this technology could stop emerging invasive pests and plant pathogen epidemics but is currently limited due to its inability to be directly applied in the field.

2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

Our work shows that using MPS is a viable method to detect known and novel pests and pathogens with or without existing genomic resources. Ultimately, this could lead to surveillance posts that collect environmental DNA and alert individuals of pests, invasive species, or pathogens detected in their area. As sequencing costs continue to decrease, these methods could become routine surveillance practices. The current hindrances in implementation are computational constraints, as MPS at high coverage

warrants high storage and memory capacities. Our findings show that pests can be reliably detected even at low relative quantities in an insect trap, as confirmed by reliable detection of Tuta absoluta in simulated traps. Another pitfall is feasibility of direct application in the field, as extensive sample processing must be undergone to prepare for MPS, and sequencing is mostly done through large sequencing facilities. More portable sequencing technology is being developed, with Oxford Nanopore technologies being the leader in this front, and this could help to eventually overcome this hurdle.

We were also able to detect novel and previously described insect pathogens; studies supported by this fund were able to detect two previously described bed bug viruses and four novel viruses associated with bed bugs. Furthermore, MPS data generated using funds from this award detected the tropical bed bug (Cimex hemipterus) in samples from Ohio, which would be the farthest North in the US this species has ever been recorded (previously, Florida was the record). This further emphasizes the capability of this technology as a method to detect potentially damaging insect pests, potentially damaging pathogens transmitted by those insects, and emerging pathogens harbored within those insects all in one assay.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is toanswer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

a) potential outcomes, i.e., findings, results, or recommendations that couldimpact U.S. agriculture and food systems, if used;
b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices,legislation, research/product design, training and so forth; and
c) c. end outcomes, i.e., how findings, results, or recommendations havecontributed to documented reductions in work-related morbidity, mortality, and/or exposure.

This project has potential outcomes that are highly relevant to U.S. agriculture. First, we demonstrate the capability of advancing MPS technology to detect potential insect threats to agriculture. As potential threats are detected, control strategies can be immediately implemented. Second, we demonstrate the capability to detect

pathogens that could be harbored by those insects. Insect pathogens are relevant in two ways: 1.) They could be pathogens of plants that are directly transmitted by the insect and stifle crop production, and 2.) They could be insect-specific pathogens that could be used in biocontrol strategies. Insect viruses have been used before as biological control agents (for example, baculoviruses in lepidopteran pests), and further characterization of the viruses that insect pests harbor is useful to develop and discover novel biological control agents. Development of this technology for farms in real-time could be highly useful to agriculture.

2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Aim 1: Develop metagenomics-based surveillance strategies for groups of pest insects and viruses of agricultural significance
We have developed computational methods to detect insect pests and known and novel viruses from DNA and RNA Illumina sequence reads. We have a manuscript in preparation that uses a metagenomic approach to detect known and novel viruses in bed bugs, and this pipeline could be easily applied to any insect of agricultural importance.

Aim 2: Further refine data analysis pipelines for efficient, locally performed, target insect and pathogen detection in raw metagenomic Oxford Nanopore and Illumina sequence datasets
We have currently only applied the analysis pipeline to Illumina sequence, although Oxford Nanopore data would be much more efficient for locally performed surveillance.

Aim 3: Develop a catalogue of North American insect pests and viral species and identify priority areas to drive future sequencing efforts towards a comprehensive, portable, EDNA or full reference library for use in agricultural settings
We have not developed a catalogue of pests and have currently only applied our efforts to some known agricultural/urban pests. These include the tomato leaf miner (*Tuta absoluta*), the redbanded stink bug (*Piezodorus guildinii*), and bed bugs (*Cimex lectularius* and *Cimex*

hemipterus).

2.5. Have any of the major goals/specific aims or milestones for the project changedsince the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No change

2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (bothpositive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Using this award, we sequenced multiple insect traps and individual insects. We confirmed that even pests of small relative size could be detected in mixed insect traps, as was the case for Tuta absoluta, a lepidopteran pest of tomatoes. We could reliably detect T. absoluta using only raw sequence reads, making the pipeline relatively simple. We were able to blindly identify Tuta-positive samples with 100% fidelity in (a) simulated insect traps (metagenomic samples composed of known insects from around the lab), (b) insect traps collected at agricultural sites, and (c) randomly spiked T. absoluta specimens in insect mixes. This is consistent with our first aim, as we were able to implement a simple computational technique to detect a relevant invasive agricultural pest.

Funds from this award were also used to sequence multiple redbanded stink bug transcriptomic samples. The redbanded stink bug is a major soybean pest that has expanded its range from South America to the southern United States and is known to have a very high damage potential. We characterized this pest's salivary transcriptome and found that many digestive protein transcripts were highly upregulated in its salivary tissues compared to the rest of the body without salivary tissues. We used its genome and eight other hemipteran genomes to find significant gene family

evolution in gene families involved in plant tissue degradation and chemical detoxification. These results shed light on its high damage potential and are useful to inform control strategies involving compounds that may be degraded by the redbanded stink bug's saliva.

We also sequenced 25 individual bed bugs from around the world and detected two known viruses and four novel viruses. This showcases our pipeline's ability to successfully detect known and novel viruses. We assume that all these viruses are bed-bug specific, and they could lead to novel host specific biocontrol agents. Our pipeline can be easily applied to other insects of agricultural importance to detect plant viruses and emerging pathogens. Additionally, these data provide evidence of the first detection of the tropical bed bug in Ohio, which shows the capability of this technology to identify emerging invasive species, known pathogens, and novel pathogens all in one assay. Furthermore, this finding was unexpected, as Florida is the northernmost US state in which a tropical bed bug has been described. This could be a sign of global warming as tropical bed bugs are mostly found in warm equatorial climates.

We were not able to implement this technology on a local scale, as all sequencing had to be outsourced. This is due in part because the technology is very difficult to work outside of a lab setting, as samples must undergo extensive processing before sequencing can be conducted. This could change in the future as portable DNA sequencing technology is becoming more feasible.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Includeapproaches taken to ensure robust and unbiased results.

Our approaches were exploratory by design, as our goal is an effective surveillance system. We kept our insect trap results unbiased by randomly assigning the Tuta spike-in and keeping the computational biologist blind to the data. Our MPS approach is inherently unbiased because it does not involve the amplification of specific genomic targets.

2.8. List key stakeholders that could be served by results of this project.

These results could be of interest to multiple economic sectors, particularly those in the agriculture sector. This technology could benefit farmers, foresters, horticulturalists, or any other field in

which insects pose a threat. Our results also extend to sectors such as pest control and pest management, homeowners, gardeners, and even health professionals as this technology could also detect emerging human disease vectors and pathogens.

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sourceswill be reported under Information Products. (up to 1,000-word limit)

Nothing to report.

2.10. Describe challenges or delays encountered during project and actions that weretaken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

During this project, employee turnover was our biggest obstacle. The primary PhD student working on this graduated, a dedicated post-doc left for a job in industry, and the PI was also transitioning out of academia at the end of this grant.

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to

500-word limit)

This project helped many students develop their computational skills. Most of them had only ever done lab-based science and only had a shallow knowledge of the methods and techniques used for MPS and MPS analysis. These students were able to able to develop skills in bioinformatics and some were able to take on lead roles in projects supported by this award, strengthening them professionally.

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reportingperiod. If other education components are involved, please describe them in detail. (up to 500-word limit)

There were three graduate students and three undergraduate students that were supported fully or in part by this grant. Another undergraduate was involved in developing computational methods as a directed individual study, and a graduate student not supported by this grant had aligned interests and assisted in the computational analyses.

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video,audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

There are two manuscripts in preparation produced from this project.

3.2. Please provide a list of citations for the information products produced during theproject.

There are no publications to cite yet, but funding will be

acknowledged at the time of their publication.

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during theproject resulting directly from the FFAR award? If yes, please provide citation.

There are currently no scholarly publications that resulted from this award.

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

There are no websites associated with this project.

3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

No patents were generated from this project.

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if theymust remain confidential indefinitely. (up to 500-word limit)

No information from this project is confidential

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers,and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access andre-use these data in the future?

All data will be uploaded to NCBI's Sequence Read Archive public database.

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation ofexisting data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

MPS data on sequenced insects was generated for this project. All raw MPS data will be uploaded to NCBI's Sequence Read Archive (SRA) upon publication. Unique SRA identifiers will be created when these are deposited into the database.

4.2. If you list multiple data sets, are these data sets related? If so, please provide ashort description of how they are related.

Each BioProject will be unrelated from each other and dependent on the focus of the sequencing projects.

4.3. Please provide copies of relevant metadata records to support FFAR's mission ofenhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Each BioProject will have descriptors of the data in the SRA database.

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Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I willnot claim otherwise.

PI Full Name: Jonas King

PI Signature: The PI has left the University and his location is unknown.

"I confirm that Dr. Jonas King has left Mississippi State University and that all attempts to contact him have failed. The report above was put together with help from his former graduate students."

Daniel G. Peterson

Digitally signed by Daniel G. Peterson
Date: 2023.04.14 15:43:23 -05'00'

Interim Department Head Date: 4/14/2023

Authorized Signing Official (ASO) Name: Tina Kinard, Associate Director Office of Sponsored Projects

ASO Signature: Tina Kinard Date: 04/21/2023